MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION DOCUMENT
49WF POTABLE WATER TREATMENT PLANT
CHEMICAL STORAGE HOUSE DESIGN

Document Number: MD-216-49WF-EG-CV-CAL-2004
Revision: D01
Revision Date: 27.10.2023
Status: AS BUILT

Outotec Document Number:	900707	GAU01	AG32	CED03	53004	OU100691545	2
L&T Document Number:	O19045	080	C	K	24373	004	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	17.10.2020	FIRST ISSUE	BJB	SJM	AG	
C01	17.11.2020	ISSUED FOR CONSTRUCTION	BJB	SJM	AG	
D01	27.10.2020	AS BUILT	BJB	SJM	AG	

TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4

2. EXECUTIVE SUMMARY 4

3. SCOPE 4

4. DESIGN REQUIREMENTS 5

- 4.1 Design 5
- 4.2 Materials 5
- 4.3 Foundations Design 5
- 4.4 Grouting 5
- 4.5 Minimum Concrete Cover 5
- 4.6 Minimum Reinforcement Steel (Rebar) Size 5
- 4.7 Allowable Bearing Capacity 6
- 4.8 Stability Against Overturning & Sliding 6

5. DESIGN LOADS 6

- 5.1 Dead Load 6
- 5.1.1 Roof Dead Load 6
- 5.1.2 Ground Floor Wall Load 8
- 5.2 Live Load 8
- 5.2.1 Roof Live Load 8

6. DESIGN CALCULATION FOR FOUNDATION 12

References 21

ATTACHMENT 1 STAAD INTPUT 22

ATTACHMENT 2 STAAD OUTPUT 26

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. INTENT

This document is to Prepare for design of **Chemical Storage House**. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the structure.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis & design of beam & column members have been carried out in STAAD. For foundation design, force values are directly taken from STAAD output & designed in excel sheet.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4. DESIGN REQUIREMENTS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar: ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundation Design

Foundations shall be isolated type. The design calculations as determined by the standard in Civil, Structural & Architectural design criteria (Doc no. MD-216-1000-EG-ST-CRT-1001).

4.4 Grouting

Grout shall be 30 mm thick cementitious under steel column bases of strength 40 Mpa.

4.5 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation Top & Column	50
Exposed to weather or contact with ground	Beam	40

4.6 Minimum Reinforcing Steel (Rebar) Size

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The minimum rebar size for use as tension, compression reinforcement shall be 12 mm, except where welded wire fabric is used. For ties and spirals the minimum rebar size shall be 10 mm.

4.7 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 750 kN/m^2 at 1.5m depth below EGL (ie. RL 917.00m) refer Table 10 - BH-21 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.8 Stability Against Overturning & Sliding

Overturning

Structures and building foundations, shall have the following minimum stability ratios against overturning:

- 1.5 for load combinations with wind loads
- 2 for other load combinations

Sliding

A minimum factor of safety against sliding shall be 1.5.

5. DESIGN LOADS

5.1 Dead Load

5.1.1 Roof Dead Load

- Provision for elecrical lighting ceiling = 0.05 kN/m^2
- Suspended steel channel system = 0.10 kN/m^2
- Acoustical Fibre Tile (16mm) = 0.05 kN/m^2

  Sub total = 0.20 kN/m^2
- Hollow Core slab (PCI 6th edition)

  (HCS-200 Topped) = 3.1 kN/m²

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Prestressed Hollowcore Floor

OVERALL STRUCTURAL DEPTH MM	SELF WT. KN/M²	NO. OF CORES IN UNIT	SPANS INDICATED BELOW ALLOW FOR CHARACTERISTIC SERVICE LOAD (LIVE LOAD) PLUS SELF WEIGHT PLUS 1.5KN/M² FOR FINISHES CHARACTERISTIC SERVICE LOADS KN/M²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.0
			Clear span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense Slab)	2.94	11	7.40	7.40	7.40	7.35	7.06	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

  100mm thk Polystyrene Foam
  0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²
- Water proofing membrane

  Bituminous, smooth surface = 0.07 kN/m²

- 90 mm thick average cement foam concrete screed Lightweight

  Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

- 75 mm Concrete screed above HCS = 21 x 0.075
  =1.6 kN/m²

Sub-total = 3.1 kN/m²
Total Load = 6.4 kN/m²

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

UDL on edge ledger beams (RB1) = 6.4 x 4.5/2 = 14.4 kN/m

Moment on all the extrenal beams due to cantilever portion of slab & Screed
= (0.15x24+(0.05+0.2)/2x20) *0.55*0.55/2 = 0.93 knm/m

5.1.2 Ground Floor Wall Load

- 250 thk. Outer CMU (Partially Grouted Cell @ 800 c-c)

  16 x 0.25 x (3.8-0.5) m clear height = 13.2 kN/m

5.2 Live Load

5.2.1 Roof Live Load

- Live loads were taken from Design Criteria clause no 4.2.2.

  Roof not accessed for maintenance = 0.5 kN/m²
  Udl Due to Live Load on the Edge Ledger Beam (RB1) = 0.5 x 4.5 / 2 kN/m
  = 1.125 kN/m

  Moment on all the External Beams due to Live Load on

- Cantilever Portion of slab = 0.5 x 0.55 x 0.55/2 = 0.075 kN/m /m

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6 WIND LOAD CALCULATION

Velocity Pressure , qz evaluated at height z shall be calculated by the following equation:

$qz = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ kN/m^2 (Refer Cl. No 7.2.8 of SBC-301)

Basic Wind Speed	V =	177	km/h
Velocity pressure exposure coefficient	Kz =	1.04	As per Table 7.2-2(Page-7/44)- z = 0 to 5m of SBC-301
Exposure Categories.	=	D	As per Project Basis of Design
Topographic Factor	K_{zt}=	1.00	As per Cl. 6.4.3.2

(As per Pt no 4 of cl 6.4.3.1 of SBC-301 , $H/L_h < 0.2$, hence topographic effect not considered)

Wind Directionality Factor	K_d =	0.85	As per Table 6.4-1(Page-6/7)
Importance factor	I =	1.00	As per Table 6.5-1(Page-6/9)
Hence, for z = 10m	qz =	1.31 Kn/m2 =	0.131 Ton/m2

Design wind pressure on structure is ; p = qh (GCpf - GCpi) (Refer Eq. 7.2-15 of SBC-301)

GCpf = External pressure co-efficient (Refer Figure No. 7.2 - 6 of SBC-301)
GCpi = Internal pressure co-efficient (Refer Figure No. 7.2 - 1 of SBC-301)

GCpi = ±0.18

WIND ALONG X DIRECTION

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = + 0.18 =	1.31 x (0.61-0.18) =	0.56	Kn/m2	(Windward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.43-0.18) =	-0.80	Kn/m2	(Leeward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.45-0.18) =	-0.83	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 2.25 m = 2.25 x 0.56 = 1.26 Kn/m

Load on Leeward columns :

Load on columns for span 2.25 m = 2.25 x (-0.80) = -1.80 Kn/m

Load on side wall columns :
Load on columns for span 2.25m = 2.25 x (-0.83) = -1.87 Kn/m

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = - 0.18 =	1.31 x (0.61+0.18) =	1.03	Kn/m2	(Windward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.43+0.18) =	-0.33	Kn/m2	(Leeward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.45+0.18) =	-0.35	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 2.25 m = 2.25 x 1.03 = 2.32 Kn/m

Load on Leeward columns :

Load on columns for span 2.25 m = 2.25 x (-0.33) = -0.74 Kn/m

Load on side wall columns :
Load on columns for span 2.25m = 2.25 x (-0.35) = -0.79 Kn/m

WIND ALONG Z DIRECTION

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = + 0.18 =	1.31 x (0.61-0.18) =	0.56	Kn/m2	(Windward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.43-0.18) =	-0.80	Kn/m2	(Leeward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.45-0.18) =	-0.83	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 2.25m = 2.25x 0.56 = 1.26 Kn/m

Load on Leeward columns :

Load on columns for span 2.25m = 2.25 x (-0.80) = -1.80 Kn/m

Load on side wall columns :

Load on columns for span 2.25 m = 2.25 x (-0.83) = -1.87 Kn/m

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)

Higher values are adopted for conservative design

Wind load for GCpi = - 0.18 =	1.31 x (0.61+0.18) =	1.03	Kn/m2	(Windward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.43+0.18) =	-0.33	Kn/m2	(Leeward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.45+0.18) =	-0.35	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 2.25m = 2.25 x (1.03) = 2.32 Kn/m

Load on Leeward columns :

Load on columns for span 2.25m = 2.25 x (-0.33) = -0.74 Kn/m

Load on side wall columns :

Load on columns for span 2.25 m = 2.25 x (-0.35) = -0.79 Kn/m

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. DESIGN CALCULATION FOR FOUNDATION

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1 (STAAD Node -23)

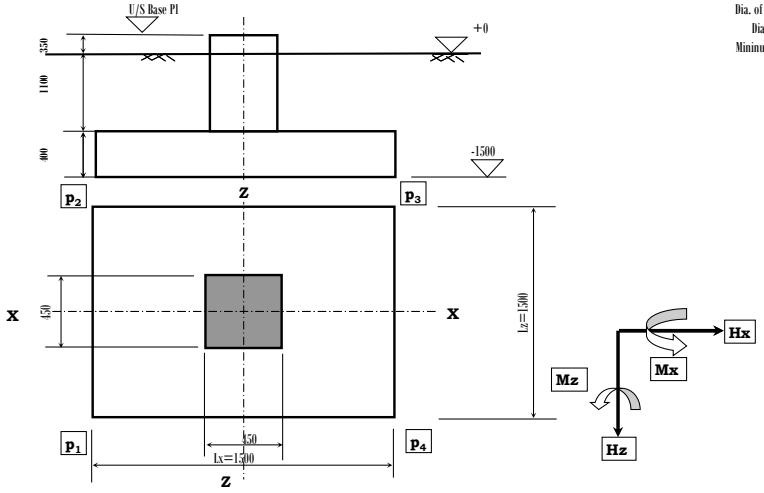
	Along-X	Along-Z
Size of column :	0.45m	0.45m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.35 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

Dead Load	DL
Fluid Load	LL
Roof Live Load	LR
Wind Load	WLX1
Wind Load	WLX2
Wind Load	WLZ1
Wind Load	WLZ2
Temp Load	TL

Calculation of Weight of Foundation :

Weight of raft & Col.= (1.5*1.5*0.4*24)+(0.45*0.45*1.45*24) = 28.65KN
Weight of soil = ((1.5*1.5)-(0.45*0.45))*1.1*18 = 40.54KN
Total = 69.19KN



Design Input:

Characteristic strength of concrete, f'_c =	35 MPa
Yield stress of steel reinforcement, f_y =	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	12mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	12mm
Dia. of bars provided along Z direction at top=	12mm
Minimum percentage of reinforcement at bottom=	0.18%

Effective depth of slab along X direction = 344mm = 0.344m
Effective depth of slab along Z direction = 332mm = 0.332m
Clear projection along Z direction = 0.525m
Clear projection along X direction = 0.525m

H_X=Force along X-X direction
Hz=Force along Z-Z direction
M_X=Moment about X-X direction
Mz=Moment about Z-Z direction

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Combined Load Cases:		**Load at top of foundation**					**Load at bottom of foundation**				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design											
201	1.0DL+1.0LL+1.0TL	143.7	11.4	0.0	8.8	0.0	212.93	11.41	21.10	8.81	16.29
202	1.0DL+1.0LR	145.5	9.9	0.0	7.1	0.0	214.66	9.91	18.34	7.11	13.15
203	1.0DL+0.75LL+0.75LR+0.75TL	145.4	11.0	0.0	8.3	0.0	214.63	11.04	20.43	8.29	15.33
204	1.0DL+1.0WLX1	136.4	3.9	0.0	6.3	0.0	205.55	3.94	7.29	6.33	11.71
205	1.0DL+1.0WLX2	149.5	16.1	0.0	6.9	0.0	218.71	16.11	29.81	6.89	12.74
206	1.0DL+1.0WLZ1	136.4	8.9	0.0	1.3	0.0	205.55	8.93	16.52	1.34	2.48
207	1.0DL+1.0WLZ2	136.4	9.5	0.0	1.9	0.0	205.55	9.49	17.55	1.88	3.49
208	1.0DL+0.75LL+0.75LR+0.75WLX1	139.9	5.4	0.0	6.4	0.0	209.09	5.44	10.07	6.43	11.90
209	1.0DL+0.75LL+0.75LR+0.75WLX2	149.8	14.6	0.0	6.8	0.0	218.97	14.57	26.96	6.85	12.67
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	139.9	9.2	0.0	2.7	0.0	209.09	9.19	16.99	2.69	4.97
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	139.9	9.6	0.0	3.1	0.0	209.09	9.60	17.76	3.10	5.73
212	0.6DL+1.0WLX1	79.2	0.0	0.0	3.4	0.0	148.37	-0.02	-0.03	3.41	6.31
213	0.6DL+1.0WLX2	92.3	12.2	0.0	4.0	0.0	161.54	12.16	22.49	3.97	7.34
214	0.6DL+1.0WLZ1	79.2	5.0	0.0	-1.6	0.0	148.37	4.97	9.20	-1.58	-2.92
215	0.6DL+1.0WLZ2	79.2	5.5	0.0	-1.0	0.0	148.37	5.53	10.23	-1.03	-1.91
Ultimate Conditions											
101	1.4DL	200.1	13.8	0.0	10.2	0.0	269.30	13.85	25.62	10.21	18.88
102	1.4DL+1.7LL+0.5LR	201.4	13.9	0.0	10.1	0.0	270.57	13.86	25.64	10.12	18.71
103	1.2DL+1.6LR+0.8WLX1	170.3	7.1	0.0	7.7	0.0	239.50	7.14	13.21	7.68	14.21
104	1.2DL+1.6LR+0.8WLX2	180.8	16.9	0.0	8.1	0.0	250.03	16.88	31.22	8.13	15.04
105	1.2DL+1.6LR+0.8WLZ1	170.3	11.1	0.0	3.7	0.0	239.50	11.13	20.59	3.69	6.83
106	1.2DL+1.6LR+0.8WLZ2	170.3	11.6	0.0	4.1	0.0	239.50	11.58	21.41	4.13	7.63
107	1.2DL+1.6LR+1.0LL	175.6	11.9	0.0	8.5	0.0	244.77	11.90	22.02	8.45	15.64
108	1.2DL+1.0LL+1.6WLX1+0.5LR	162.3	2.4	0.0	7.1	0.0	231.45	2.36	4.36	7.12	13.17
109	1.2DL+1.0LL+1.6WLX1+0.5LR	183.3	21.8	0.0	8.0	0.0	252.52	21.83	40.39	8.01	14.81
110	1.2DL+1.0LL+1.6WLX1+0.5LR	162.3	10.3	0.0	-0.9	0.0	231.45	10.34	19.13	-0.87	-1.60
111	1.2DL+1.0LL+1.6WLX1+0.5LR	162.3	11.2	0.0	0.0	0.0	231.45	11.23	20.78	0.01	0.01
112	1.2DL+1.0LL	171.5	11.9	0.0	8.7	0.0	240.72	11.87	21.96	8.75	16.19
113	0.9DL+1.6WLX1	118.1	-0.6	0.0	5.0	0.0	187.30	-0.62	-1.15	5.02	9.29
114	0.9DL+1.6WLX2	139.2	18.9	0.0	5.9	0.0	208.37	18.86	34.88	5.91	10.94
115	0.9DL+1.6WLZ1	118.1	7.4	0.0	-3.0	0.0	187.30	7.36	13.62	-2.96	-5.48
116	0.9DL+1.6WLZ2	118.1	8.3	0.0	-2.1	0.0	187.30	8.25	15.27	-2.09	-3.87
117	1.4DL+ 1.4TL+1.7LL+0.5LR	202.5	16.0	0.0	12.2	0.0	271.70	15.98	29.56	12.23	22.63

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0LL+1.0TL	212.9	11.4	21.1	8.8	16.3	7.6	9.8	7.5	9.7
202	1.0DL+1.0LR	214.7	9.9	18.3	7.1	13.1	8.8	12.2	8.7	12.1
203	1.0DL+0.75LL+0.75LR+0.75TL	214.6	11.0	20.4	8.3	15.3	7.9	10.5	7.8	10.4
204	1.0DL+1.0WLX1	205.5	3.9	7.3	6.3	11.7	21.1	13.2	20.9	13.0
205	1.0DL+1.0WLX2	218.7	16.1	29.8	6.9	12.7	5.5	12.9	5.4	12.7
206	1.0DL+1.0WLZ1	205.5	8.9	16.5	1.3	2.5	9.3	62.2	9.2	61.4
207	1.0DL+1.0WLZ2	205.5	9.5	17.5	1.9	3.5	8.8	44.2	8.7	43.6
208	1.0DL+0.75LL+0.75LR+0.75WLX1	209.1	5.4	10.1	6.4	11.9	15.6	13.2	15.4	13.0
209	1.0DL+0.75LL+0.75LR+0.75WLX2	219.0	14.6	27.0	6.8	12.7	6.1	13.0	6.0	12.8
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	209.1	9.2	17.0	2.7	5.0	9.2	31.5	9.1	31.1
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	209.1	9.6	17.8	3.1	5.7	8.8	27.4	8.7	27.0
212	0.6DL+1.0WLX1	148.4	0.0	0.0	3.4	6.3	3538	17.6	3491	17.4
213	0.6DL+1.0WLX2	161.5	12.2	22.5	4.0	7.3	5.4	16.5	5.3	16.3
214	0.6DL+1.0WLZ1	148.4	5.0	9.2	-1.6	-2.9	12.1	38.1	11.9	37.6
215	0.6DL+1.0WLZ2	148.4	5.5	10.2	-1.0	-1.9	10.9	58.2	10.7	57.5
						Min.=	5.4	9.8	5.3	9.7
							OK	OK	OK	OK

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Load Combination & Calculation of Base pressure

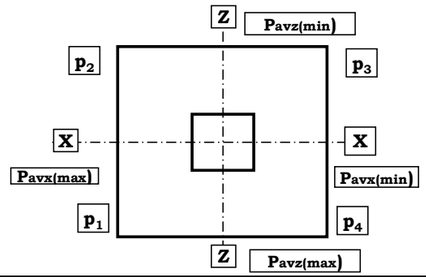
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 500 kN/m^2 For WL/SL case = 625 kN/m^2

Gross SBC of soil for DL, LL= 527 kN/m^2 For WL/SL case = 652 kN/m^2



Allowable Stress Design

Load Case	Load Combination	Design Load/moments V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ (Lx)^2.Lz	6.Mx/ Lx.(Lz)^2	Base Pressure (kN/m^2) p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-γh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0LL+1.0TL	212.93	21.10	16.29	94.64	37.51	28.96	161.11	103.19	86.08	28.17	134.11	132.15	57.13	123.60	65.68	O.K
202	1.0DL+1.0LR	214.66	18.34	13.15	95.40	32.60	23.37	151.37	104.63	86.18	39.44	124.37	128.00	62.81	118.78	72.04	O.K
203	1.0DL+0.75LL+0.75LR+0.75TL	214.63	20.43	15.33	95.39	36.31	27.26	158.96	104.45	86.33	31.82	131.96	131.71	59.08	122.65	68.14	O.K
204	1.0DL+1.0WLX1	205.55	7.29	11.71	91.35	12.96	20.82	125.13	83.50	99.21	57.58	98.13	104.32	78.40	112.17	70.54	O.K
205	1.0DL+1.0WLX2	218.71	29.81	12.74	97.21	52.99	22.64	172.84	127.55	66.86	21.57	145.84	150.20	44.22	119.85	74.56	O.K
206	1.0DL+1.0WLZ1	205.55	16.52	2.48	91.35	29.37	4.40	125.13	116.32	66.38	57.58	98.13	120.73	61.98	95.76	86.95	O.K
207	1.0DL+1.0WLZ2	205.55	17.55	3.49	91.35	31.20	6.20	128.75	116.36	66.35	53.96	101.75	122.56	60.16	97.55	85.16	O.K
208	1.0DL+0.75LL+0.75LR+0.75WLX	209.09	10.07	11.90	92.93	17.90	21.15	131.98	89.68	96.18	53.88	104.98	110.83	75.03	114.08	71.78	O.K
209	1.0DL+0.75LL+0.75LR+0.75WLX	218.97	26.96	12.67	97.32	47.92	22.52	167.76	122.72	71.91	26.88	140.76	145.24	49.40	119.84	74.80	O.K
210	1.0DL+0.75LL+0.75LR+0.75WLZ	209.09	16.99	4.97	92.93	30.21	8.84	131.98	114.30	71.56	53.88	104.98	123.14	62.72	101.77	84.09	O.K
211	1.0DL+0.75LL+0.75LR+0.75WLZ	209.09	17.76	5.73	92.93	31.58	10.19	134.69	114.32	71.54	51.17	107.69	124.51	61.36	103.12	82.75	O.K
212	0.6DL+1.0WLX1	148.37	-0.03	6.31	65.94	0.06	11.22	77.22	54.77	77.11	54.66	50.22	66.00	65.89	77.17	54.72	O.K
213	0.6DL+1.0WLX2	161.54	22.49	7.34	71.79	39.98	13.05	124.82	98.72	44.87	18.77	97.82	111.77	31.82	84.85	58.75	O.K
214	0.6DL+1.0WLZ1	148.37	9.20	-2.92	65.94	16.36	5.19	87.49	77.11	54.77	44.39	60.49	82.30	49.58	71.13	60.75	O.K
215	0.6DL+1.0WLZ2	148.37	10.23	-1.91	65.94	18.18	3.40	87.52	80.73	51.16	44.36	60.52	84.13	47.76	69.34	62.55	O.K

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Ultimate Conditions

101 1.4DL	269.30	25.62	18.88	119.69	45.55	33.57	198.81	131.67	107.71	40.57	171.81	165.24	74.14	153.26	86.12	
102 1.4DL+1.7LL+0.5LR	270.57	25.64	18.71	120.25	45.58	33.27	199.10	132.57	107.94	41.41	172.10	165.84	74.68	153.52	86.99	
103 1.2DL+1.6LR+0.8WLX1	239.50	13.21	14.21	106.44	23.48	25.27	155.19	104.65	108.24	57.70	128.19	129.92	82.97	131.72	81.18	
104 1.2DL+1.6LR+0.8WLX2	250.03	31.22	15.04	111.13	55.50	26.73	193.36	139.90	82.35	28.89	166.36	166.63	55.62	137.86	84.40	
105 1.2DL+1.6LR+0.8WLZ1	239.50	20.59	6.83	106.44	36.61	12.14	155.19	130.91	81.98	57.70	128.19	143.05	69.84	118.59	94.31	
106 1.2DL+1.6LR+0.8WLZ2	239.50	21.41	7.63	106.44	38.07	13.57	158.09	130.94	81.95	54.80	131.09	144.52	68.38	120.02	92.87	
107 1.2DL+1.6LR+1.0LL	244.77	22.02	15.64	108.79	39.14	27.80	175.72	120.12	97.45	41.85	148.72	147.92	69.65	136.59	80.99	
108 1.2DL+1.0LL+1.6WLX1+0.5LR	231.45	4.36	13.17	102.87	7.75	23.41	134.03	87.20	118.53	71.71	107.03	110.62	95.12	126.28	79.46	
109 1.2DL+1.0LL+1.6WLX1+0.5LR	252.52	40.39	14.81	112.23	71.80	26.33	210.36	157.70	66.76	14.10	183.36	184.03	40.43	138.56	85.90	
110 1.2DL+1.0LL+1.6WLX1+0.5LR	231.45	19.13	-1.60	102.87	34.01	2.85	139.73	134.03	71.71	66.00	112.73	136.88	68.86	105.72	100.02	
111 1.2DL+1.0LL+1.6WLX1+0.5LR	231.45	20.78	0.01	102.87	36.93	0.02	139.82	139.78	65.95	65.92	112.82	139.80	65.94	102.89	102.85	
112 1.2DL+1.0LL	240.72	21.96	16.19	106.99	39.04	28.77	174.80	117.25	96.72	39.17	147.80	146.03	67.95	135.76	78.21	
113 0.9DL+1.6WLX1	187.30	-1.15	9.29	83.25	2.04	16.52	101.80	68.76	97.73	64.69	74.80	85.28	81.21	99.77	66.73	
114 0.9DL+1.6WLX2	208.37	34.88	10.94	92.61	62.01	19.44	174.06	135.18	50.04	11.15	147.06	154.62	30.60	112.05	73.17	
115 0.9DL+1.6WLZ1	187.30	13.62	-5.48	83.25	24.22	9.74	117.21	97.72	68.77	49.28	90.21	107.47	59.03	92.99	73.50	
116 0.9DL+1.6WLZ2	187.30	15.27	-3.87	83.25	27.14	6.87	117.26	103.51	62.98	49.23	90.26	110.39	56.11	90.12	76.37	
117 1.4DL+ 1.4TL+1.7LL+0.5LR	271.70	29.56	22.63	120.75	52.55	40.24	213.54	133.07	108.44	27.97	186.54	173.31	68.21	160.99	80.52	
Max :	219	30	16													
Min :	148	0	-3													

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 134.11 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 144.11 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 145.84 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 155.84 kN/m²

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0LR

Maximum pressure, p_1 (pmax) = 95.40+32.60+23.37= 151.37 kN/m²

p_2= 95.40+32.60-23.37= 104.63 kN/m²

p_3= 95.40-(32.60+23.37)= 39.44 kN/m²

p_4= 95.40-32.60+23.37= 86.18 kN/m²

Maxm pressure excluding weight of soil (p_{max}-γh)= 151.37-1.5*18= 124.37 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (151.37+104.63)/2 = 128.00 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (39.44+86.18)/2= 62.81 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (151.37+86.18)/2= 118.78 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (39.44+104.63)/2= 72.03 kN/m²

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

DESIGN OF FOUNDATION MKD. F1

Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.45+2*0.5*0.332)+(0.45+2*0.5*0.332))*2

= 3.128 m

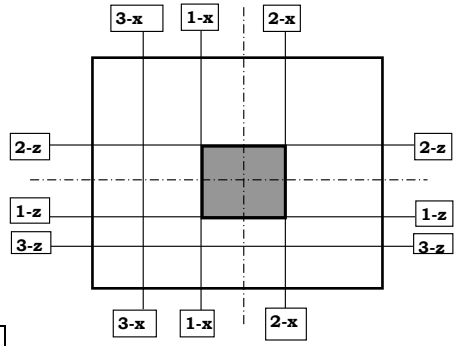
Punching Area= ((0.45+2*0.5*0.332)*(0.45+2*0.5*0.332))

= 0.61 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)

= 29.40 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M_{1x} kNm	B.M_{2x} kNm	S.F_{3x} kN	S.$F_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL	133.36	106.03	154.25	17.26	7.63	23.59	148.08
102	1.4DL+1.7LL+0.5LR	133.93	106.59	154.84	17.34	7.71	23.70	149.00
103	1.2DL+1.6LR+0.8WLX1	113.49	99.40	124.25	13.10	8.14	17.68	126.35
104	1.2DL+1.6LR+0.8WLX2	127.78	94.47	153.23	17.13	5.40	23.63	134.03
105	1.2DL+1.6LR+0.8WLZ1	117.43	95.46	134.22	14.49	6.75	19.77	126.35
106	1.2DL+1.6LR+0.8WLZ2	117.87	95.03	135.33	14.64	6.60	20.01	126.35
107	1.2DL+1.6LR+1.0LL	120.53	97.04	138.48	15.08	6.81	20.60	130.19
108	1.2DL+1.0LL+1.6WLX1+0.5LR	105.20	100.55	108.75	10.94	9.31	14.53	120.48
109	1.2DL+1.0LL+1.6WLX1+0.5LR	133.77	90.69	166.70	19.00	3.83	26.42	135.84
110	1.2DL+1.0LL+1.6WLX1+0.5LR	113.07	92.67	128.67	13.72	6.53	18.71	120.48
111	1.2DL+1.0LL+1.6WLX1+0.5LR	113.95	91.79	130.89	14.03	6.22	19.18	120.48
112	1.2DL+1.0LL	118.70	95.28	136.61	14.82	6.57	20.26	127.24
113	0.9DL+1.6WLX1	83.86	82.63	84.79	7.64	7.21	10.07	88.31
114	0.9DL+1.6WLX2	111.21	74.01	139.65	15.26	2.16	21.31	103.66
115	0.9DL+1.6WLZ1	90.52	75.98	101.62	9.98	4.86	13.60	88.31
116	0.9DL+1.6WLZ2	91.39	75.11	103.84	10.29	4.55	14.07	88.31
117	1.4DL+ 1.4TL+1.7LL+0.5LR	136.53	105.00	160.63	18.14	7.04	24.90	149.82

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

p_{1x} = 165.24-(165.24-74.14)*0.525/1.5 = 133.36 kN/m^2 Overhang = 0.525 m

p_{2x} = 74.14+(165.24-74.14)*0.525/1.5 = 106.03 kN/m^2 Effective depth = 0.344 m

p_{3x} = 165.24-(165.24-74.14)*(0.525-0.344)/1.5 = 154.25 kN/m^2

BM_{1x} = (133.36-29.4)*0.525*0.525/2+0.5*(165.24-133.36)*0.525*0.525*0.67 = 17.26 kN-m/m

BM_{2x} = (106.03-29.4)*0.525*0.525/2-0.5*(-74.14+106.03)*0.525*0.525*0.67 = 7.63 kN-m/m

SF_{3x} = (154.25-29.4)*(0.525-0.344)+0.5*(165.24-154.25)*(0.525-0.344) = 23.59 kN/m

$S.F_{(punching)}$ = (2.25-0.61)*(119.69-29.4) = 148.08 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	19.00	0.16	0.21	0.001	175.99	619.20	12ϕT	182.65	300	175	646.27

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc)/bd = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 26.42 kN/m

Shear stress , τ_v= (26.42*1000)/(1000*344)

= 0.08 N/mm^2 **O.K.**

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 149.82 kN

Strength of Concrete in shear Vc1 = 3182.95 kN **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 2227.79 kN **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 2121.96 kN **Table 22.6.5.2_ (c)**

Design Vc= 2121.96 kN

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (149.82*1000)/(3128*344)

= 0.14 N/mm^2 **No shear reinforcement reqd.**

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

p3z =pressure at a distance d from the face of column at section 3z-3z in kN/m^2.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL	129.76	109.62	144.62	16.00	8.89	23.07
102	1.4DL+1.7LL+0.5LR	130.23	110.28	144.96	16.05	9.00	23.13
103	1.2DL+1.6LR+0.8WLX1	114.03	98.87	125.22	13.30	7.94	19.12
104	1.2DL+1.6LR+0.8WLX2	119.15	103.11	130.98	14.10	8.43	20.27
105	1.2DL+1.6LR+0.8WLZ1	110.09	102.81	115.47	11.91	9.33	16.91
106	1.2DL+1.6LR+0.8WLZ2	110.52	102.37	116.53	12.06	9.18	17.15
107	1.2DL+1.6LR+1.0LL	117.13	100.45	129.44	13.89	7.99	20.00
108	1.2DL+1.0LL+1.6WLX1+0.5LR	109.89	95.85	120.26	12.61	7.64	18.12
109	1.2DL+1.0LL+1.6WLX1+0.5LR	120.13	104.33	131.78	14.21	8.62	20.41
110	1.2DL+1.0LL+1.6WLX1+0.5LR	103.73	102.02	104.99	10.43	9.82	14.66
111	1.2DL+1.0LL+1.6WLX1+0.5LR	102.88	102.86	102.88	10.13	10.12	14.18
112	1.2DL+1.0LL	115.62	98.35	128.36	13.74	7.64	19.81
113	0.9DL+1.6WLX1	88.21	78.29	95.52	9.17	5.67	13.17
114	0.9DL+1.6WLX2	98.44	86.78	107.05	10.77	6.65	15.47
115	0.9DL+1.6WLZ1	86.17	80.32	90.48	8.45	6.39	12.03
116	0.9DL+1.6WLZ2	85.31	81.18	88.35	8.15	6.69	11.55
117	1.4DL+ 1.4TL+1.7LL+0.5LR	132.83	108.68	150.64	16.85	8.33	24.40

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Sample calculation of BM & SF

Load case : 101 1.4DL

p_{1z} = 153.26-(153.26-86.12)*0.525/1.5= 129.76 kN/m^2

p_{2z} = 86.12+(153.26-86.12)*0.525/1.5= 109.62 kN/m^2

p_{3z} = 153.26-(153.26-86.12)*(0.525-0.332)/1.5= 144.62 kN/m^2

Overhang = 0.525 m

Effective depth = 0.332 m

BM_{1z} = (129.76-29.4)*0.525*0.525/2+0.5*(153.26-129.76)*0.525*0.525*0.67 = 16.00 kN-m/m

BM_{2z} = (109.62-29.4)*0.525*0.525/2-0.5*(-86.12+109.62)*0.525*0.525*0.67 = 8.89 kN-m/m

SF_{3z} = (144.62-29.4)*(0.525-0.332)+0.5*(153.26-144.62)*(0.525-0.332) = 23.07 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	16.85	0.153	0.204	0.0005	167.56	597.60	12ϕT	189.25	300	175	646.3

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 24.40 kN/m

Shear stress , τ_v= (24.4*1000)/(1000*332)

= 0.07 N/mm^2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **175** c/c

At Top **12ϕT** **175** c/c

Along Z Direction :

At Bottom **12ϕT** **175** c/c

At Top **12ϕT** **175** c/c

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ATTACHMENT 1 STAAD INPUT

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 13.10.2020
ENGINEER NAME BJB
CHECKER NAME SJM
APPROVED NAME AG
JOB CLIENT MAA'DEN
JOB REV 0
JOB NO O19045
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 2 4.50001 0 0; 4 0 3.79961 0; 5 4.50001 3.79961 0; 7 0 0 4.50001;
10 0 3.79961 4.50001; 23 0 -2 0; 24 4.50001 -2 0; 26 0 -2 4.50001;
79 4.5 0 4.50001; 80 4.5 3.79961 4.50001; 81 4.5 -2 4.50001;
MEMBER INCIDENCES
1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;
34 7 26; 583 79 80; 584 79 2; 585 79 81; 587 5 80; 589 7 79; 590 10 80;
START GROUP DEFINITION
MEMBER
_COLUMN_C1 3 4 8 31 32 34 583 585
_RB_1 11 587
_RB_2 590
_GB_1 25 30 584 589
_RB_4 1
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
CONSTANTS
MATERIAL CONCRETE ALL
MEMBER PROPERTY AMERICAN
3 4 8 31 32 34 583 585 PRIS YD 0.45 ZD 0.45
1 590 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
25 30 584 589 PRIS YD 0.6 ZD 0.25
MEMBER PROPERTY AMERICAN
11 587 PRIS YD 0.5 ZD 0.25
SUPPORTS
23 24 26 81 FIXED
*************************************************************************************************
****
LOAD 1 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 30 584 589 UNI GY -13.2
11 587 UNI GY -14.4
11 UMOM GZ 0.93
587 UMOM GZ -0.93
1 UMOM GX -0.93
590 UMOM GX 0.93
*************************************************************************************************
****
LOAD 2 LOADTYPE Live  TITLE LL
* LOAD DUE TO MONORAIL
*************************************************************************************************
****
LOAD 3 LOADTYPE Roof Live  TITLE LR
MEMBER LOAD
11 587 UNI GY -1.125
11 UMOM GZ 0.075
587 UMOM GZ -0.075
1 UMOM GX -0.075
590 UMOM GX 0.075
*************************************************************************************************
****
LOAD 4 LOADTYPE Wind  TITLE WIND ALONG X DIR. ( CPI = + 0.18) WLX1
MEMBER LOAD
3 8 UNI GX 1.26
4 583 UNI GX 1.8
8 583 UNI GZ 1.87
3 4 UNI GZ -1.87
*************************************************************************************************
****
LOAD 5 LOADTYPE Wind  TITLE WIND ALONG X DIR. ( CPI = - 0.18) WLX2
MEMBER LOAD
```

```
LOAD 6 LOADTYPE Wind  TITLE WIND ALONG Z DIR. ( CPI = + 0.18) WLZ1
MEMBER LOAD
3 UNI GX -1.87
8 UNI GX -1.87
4 UNI GX 1.87
583 UNI GX 1.87
3 4 UNI GZ 1.26
8 583 UNI GZ 1.8
*************************************************************************************************************
****
LOAD 7 LOADTYPE Wind  TITLE WIND ALONG Z DIR. ( CPI = - 0.18) WLZ2
MEMBER LOAD
3 UNI GX -0.79
8 UNI GX -0.79
4 UNI GX 0.79
583 UNI GX 0.79
3 4 UNI GZ 2.32
8 583 UNI GZ 0.74
*************************************************************************************************************
****
LOAD 8 LOADTYPE Temperature  TITLE TEMPERATURE LOAD ( TL )
TEMPERATURE LOAD
3 4 8 583 TEMP 25
1 11 587 590 TEMP 0 25
3 4 8 583 TEMP 0 0 25
*************************************************************************************************************
****
* FACTORED LOAD COMBINATIONS
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7LL+0.5LR
1 1.4 2 1.7 3 0.5
LOAD COMB 103 1.2DL+1.6LR+0.8WLX1
1 1.2 3 1.6 4 0.8
LOAD COMB 104 1.2DL+1.6LR+0.8WLX2
1 1.2 3 1.6 5 0.8
LOAD COMB 105 1.2DL+1.6LR+0.8WLZ1
1 1.2 3 1.6 6 0.8
LOAD COMB 106 1.2DL+1.6LR+0.8WLZ2
1 1.2 3 1.6 7 0.8
LOAD COMB 107 1.2DL+1.6LR+1.0LL
1 1.2 2 1.0 3 1.6
LOAD COMB 108 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 4 1.6
LOAD COMB 109 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 5 1.6
LOAD COMB 110 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 6 1.6
LOAD COMB 111 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 7 1.6
LOAD COMB 112 1.2DL+1.0LL
1 1.2 2 1.0
LOAD COMB 113 0.9DL+1.6WLX1
1 0.9 4 1.6
LOAD COMB 114 0.9DL+1.6WLX2
1 0.9 5 1.6
LOAD COMB 115 0.9DL+1.6WLZ1
1 0.9 6 1.6
LOAD COMB 116 0.9DL+1.6WLZ2
1 0.9 7 1.6
LOAD COMB 117 1.4DL+ 1.4TL+1.7LL+0.5LR
1 1.4 2 1.7 3 0.5 8 1.4
*************************************************************************************************************
****
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
LOAD COMB 201 1.0DL+1.0LL+1.0TL
1 1.0 2 1.0 8 1.0
LOAD COMB 202 1.0DL+1.0LR
1 1.0 3 1.0
LOAD COMB 203 1.0DL+0.75LL+0.75LR+0.75TL
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 204 1.0DL+1.0WLX1
1 1.0 4 1.0
LOAD COMB 205 1.0DL+1.0WLX2
1 1.0 5 1.0
LOAD COMB 206 1.0DL+1.0WLZ1
1 1.0 6 1.0
LOAD COMB 207 1.0DL+1.0WLZ2
1 1.0 7 1.0
LOAD COMB 208 1.0DL+0.75LL+0.75LR+0.75WLX1
1 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 209 1.0DL+0.75LL+0.75LR+0.75WLX2
1 1.0 2 0.75 3 0.75 5 0.75
LOAD COMB 210 1.0DL+0.75LL+0.75LR+0.75WLZ1
```

```
1 0.6 4 1.0
LOAD COMB 213 0.6DL+1.0WLX2
1 0.6 5 1.0
LOAD COMB 214 0.6DL+1.0WLZ1
1 0.6 6 1.0
LOAD COMB 215 0.6DL+1.0WLZ2
1 0.6 7 1.0
*******************************************************************************************************************
****
PERFORM ANALYSIS
LOAD LIST 101 TO 117
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MAXMAIN 20 MEMB 3 4 8 31 32 34 583 585
SFACE 0.5 MEMB 1 11 25 30 584 587 589 590
MINMAIN 20 MEMB 3 4 8 31 32 34 583 585
MAXMAIN 20 MEMB 1 11 587 590
MINMAIN 12 MEMB 1 11 587 590
MINMAIN 16 MEMB 25 30 584 589
MINSEC 10 ALL
DESIGN BEAM 1 11 25 30 584 587 589 590
DESIGN COLUMN 3 4 8 31 32 34 583 585
CONCRETE TAKE
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 2 STAAD OUTPUT

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6           *
*           Version  20.07.11.70                  *
*           Proprietary Program of                *
*           Bentley Systems, Inc.                 *
*           Date=    OCT 16, 2020                 *
*           Time=    21:56:25                     *
*                                                *
*      USER ID: LNT ECC DIV                      *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\MAADEN\Potable Water Treatment Plant\Chemical Storage\STAAD\Chemical Storage_PWTP.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 13.10.2020
    4. ENGINEER NAME BJB
    5. CHECKER NAME SJM
    6. APPROVED NAME AG
    7. JOB CLIENT MAA'DEN
    8. JOB REV 0
    9. JOB NO O19045
   10. END JOB INFORMATION
   11. INPUT WIDTH 79
   12. UNIT METER KN
   13. JOINT COORDINATES
   14. 1 0 0 0; 2 4.50001 0 0; 4 0 3.79961 0; 5 4.50001 3.79961 0; 7 0 0 4.50001
   15. 10 0 3.79961 4.50001; 23 0 -2 0; 24 4.50001 -2 0; 26 0 -2 4.50001
   16. 79 4.5 0 4.50001; 80 4.5 3.79961 4.50001; 81 4.5 -2 4.50001
   17. MEMBER INCIDENCES
   18. 1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24
   19. 34 7 26; 583 79 80; 584 79 2; 585 79 81; 587 5 80; 589 7 79; 590 10 80
   20. START GROUP DEFINITION
   21. MEMBER
   22. _COLUMN_C1 3 4 8 31 32 34 583 585
   23. _RB_1 11 587
   24. _RB_2 590
   25. _GB_1 25 30 584 589
   26. _RB_4 1
   27. END GROUP DEFINITION
   28. DEFINE MATERIAL START
   29. ISOTROPIC CONCRETE
   30. E 2.17184E+007
   31. POISSON 0.17
   32. DENSITY 23.6158
   33. ALPHA 5E-006
   34. DAMP 0.05
   35. TYPE CONCRETE
   36. STRENGTH FCU 27578.9
   37. END DEFINE MATERIAL
   38. CONSTANTS
```

```
39. MATERIAL CONCRETE ALL
40. MEMBER PROPERTY AMERICAN
41. 3 4 8 31 32 34 583 585 PRIS YD 0.45 ZD 0.45
42. 1 590 PRIS YD 0.5 ZD 0.25
43. MEMBER PROPERTY AMERICAN
44. 25 30 584 589 PRIS YD 0.6 ZD 0.25
45. MEMBER PROPERTY AMERICAN
46. 11 587 PRIS YD 0.5 ZD 0.25
47. SUPPORTS
48. 23 24 26 81 FIXED
49. ******************************************************************************
50. LOAD 1 LOADTYPE DEAD  TITLE DL
51. SELFWEIGHT Y -1
52. MEMBER LOAD
53. 25 30 584 589 UNI GY -13.2
54. 11 587 UNI GY -14.4
55. 11 UMOM GZ 0.93
56. 587 UMOM GZ -0.93
57. 1 UMOM GX -0.93
58. 590 UMOM GX 0.93
59. ******************************************************************************
60. LOAD 2 LOADTYPE LIVE  TITLE LL
61. * LOAD DUE TO MONORAIL
62. ******************************************************************************
63. LOAD 3 LOADTYPE ROOF LIVE  TITLE LR
64. MEMBER LOAD
65. 11 587 UNI GY -1.125
66. 11 UMOM GZ 0.075
67. 587 UMOM GZ -0.075
68. 1 UMOM GX -0.075
69. 590 UMOM GX 0.075
70. ******************************************************************************
71. LOAD 4 LOADTYPE WIND  TITLE WIND ALONG X DIR. ( CPI = + 0.18) WLX1
72. MEMBER LOAD
73. 3 8 UNI GX 1.26
74. 4 583 UNI GX 1.8
75. 8 583 UNI GZ 1.87
76. 3 4 UNI GZ -1.87
77. ******************************************************************************
78. LOAD 5 LOADTYPE WIND  TITLE WIND ALONG X DIR. ( CPI = - 0.18) WLX2
79. MEMBER LOAD
80. 3 8 UNI GX -0.74
81. 4 583 UNI GX -2.32
82. 8 583 UNI GZ 0.79
83. 3 4 UNI GZ -0.79
84. ******************************************************************************
85. LOAD 6 LOADTYPE WIND  TITLE WIND ALONG Z DIR. ( CPI = + 0.18) WLZ1
86. MEMBER LOAD
87. 3 UNI GX -1.87
88. 8 UNI GX -1.87
89. 4 UNI GX 1.87
90. 583 UNI GX 1.87
91. 3 4 UNI GZ 1.26
92. 8 583 UNI GZ 1.8
93. ******************************************************************************
94. LOAD 7 LOADTYPE WIND  TITLE WIND ALONG Z DIR. ( CPI = - 0.18) WLZ2
```

```
 95. MEMBER LOAD
 96. 3 UNI GX -0.79
 97. 8 UNI GX -0.79
 98. 4 UNI GX 0.79
 99. 583 UNI GX 0.79
100. 3 4 UNI GZ 2.32
101. 8 583 UNI GZ 0.74
102. ********************************************************************************
103. LOAD 8 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD ( TL )
104. TEMPERATURE LOAD
105. 3 4 8 583 TEMP 25
106. 1 11 587 590 TEMP 0 25
107. 3 4 8 583 TEMP 0 0 25
108. ********************************************************************************
109. * FACTORED LOAD COMBINATIONS
110. LOAD COMB 101 1.4DL
111. 1 1.4
112. LOAD COMB 102 1.4DL+1.7LL+0.5LR
113. 1 1.4 2 1.7 3 0.5
114. LOAD COMB 103 1.2DL+1.6LR+0.8WLX1
115. 1 1.2 3 1.6 4 0.8
116. LOAD COMB 104 1.2DL+1.6LR+0.8WLX2
117. 1 1.2 3 1.6 5 0.8
118. LOAD COMB 105 1.2DL+1.6LR+0.8WLZ1
119. 1 1.2 3 1.6 6 0.8
120. LOAD COMB 106 1.2DL+1.6LR+0.8WLZ2
121. 1 1.2 3 1.6 7 0.8
122. LOAD COMB 107 1.2DL+1.6LR+1.0LL
123. 1 1.2 2 1.0 3 1.6
124. LOAD COMB 108 1.2DL+1.0LL+1.6WLX1+0.5LR
125. 1 1.2 2 1.0 3 0.5 4 1.6
126. LOAD COMB 109 1.2DL+1.0LL+1.6WLX1+0.5LR
127. 1 1.2 2 1.0 3 0.5 5 1.6
128. LOAD COMB 110 1.2DL+1.0LL+1.6WLX1+0.5LR
129. 1 1.2 2 1.0 3 0.5 6 1.6
130. LOAD COMB 111 1.2DL+1.0LL+1.6WLX1+0.5LR
131. 1 1.2 2 1.0 3 0.5 7 1.6
132. LOAD COMB 112 1.2DL+1.0LL
133. 1 1.2 2 1.0
134. LOAD COMB 113 0.9DL+1.6WLX1
135. 1 0.9 4 1.6
136. LOAD COMB 114 0.9DL+1.6WLX2
137. 1 0.9 5 1.6
138. LOAD COMB 115 0.9DL+1.6WLZ1
139. 1 0.9 6 1.6
140. LOAD COMB 116 0.9DL+1.6WLZ2
141. 1 0.9 7 1.6
142. LOAD COMB 117 1.4DL+ 1.4TL+1.7LL+0.5LR
143. 1 1.4 2 1.7 3 0.5 8 1.4
144. ********************************************************************************
145. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
146. LOAD COMB 201 1.0DL+1.0LL+1.0TL
147. 1 1.0 2 1.0 8 1.0
148. LOAD COMB 202 1.0DL+1.0LR
149. 1 1.0 3 1.0
150. LOAD COMB 203 1.0DL+0.75LL+0.75LR+0.75TL
```

```
151. 1 1.0 2 0.75 3 0.75 8 0.75
152. LOAD COMB 204 1.0DL+1.0WLX1
153. 1 1.0 4 1.0
154. LOAD COMB 205 1.0DL+1.0WLX2
155. 1 1.0 5 1.0
156. LOAD COMB 206 1.0DL+1.0WLZ1
157. 1 1.0 6 1.0
158. LOAD COMB 207 1.0DL+1.0WLZ2
159. 1 1.0 7 1.0
160. LOAD COMB 208 1.0DL+0.75LL+0.75LR+0.75WLX1
161. 1 1.0 2 0.75 3 0.75 4 0.75
162. LOAD COMB 209 1.0DL+0.75LL+0.75LR+0.75WLX2
163. 1 1.0 2 0.75 3 0.75 5 0.75
164. LOAD COMB 210 1.0DL+0.75LL+0.75LR+0.75WLZ1
165. 1 1.0 2 0.75 3 0.75 6 0.75
166. LOAD COMB 211 1.0DL+0.75LL+0.75LR+0.75WLZ2
167. 1 1.0 2 0.75 3 0.75 7 0.75
168. LOAD COMB 212 0.6DL+1.0WLX1
169. 1 0.6 4 1.0
170. LOAD COMB 213 0.6DL+1.0WLX2
171. 1 0.6 5 1.0
172. LOAD COMB 214 0.6DL+1.0WLZ1
173. 1 0.6 6 1.0
174. LOAD COMB 215 0.6DL+1.0WLZ2
175. 1 0.6 7 1.0
176. ********************************************************************************
177. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**12**	**NUMBER OF MEMBERS**	**16**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**4**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 8/ 5/ 30 DOF
TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 48
TOTAL LOAD COMBINATION CASES = 32 SO FAR.
SIZE OF STIFFNESS MATRIX = 2 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.0/ 323626.9 MB

```
178. LOAD LIST 101 TO 117
179. START CONCRETE DESIGN
180. CODE ACI
181. TRACK 2 ALL
```

```
182. FC 35000 ALL
183. CLB 0.04 ALL
184. CLS 0.04 ALL
185. CLT 0.04 ALL
186. MAXMAIN 20 MEMB 3 4 8 31 32 34 583 585
187. SFACE 0.5 MEMB 1 11 25 30 584 587 589 590
188. MINMAIN 20 MEMB 3 4 8 31 32 34 583 585
189. MAXMAIN 20 MEMB 1 11 587 590
190. MINMAIN 12 MEMB 1 11 587 590
191. MINMAIN 16 MEMB 25 30 584 589
192. MINSEC 10 ALL
193. DESIGN BEAM 1 11 25 30 584 587 589 590
```

ACI 318-11 BEAM NO. 1 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 17.48 KN-MET AT 2250.MM, LOAD 117					
	REQD STEEL= 142.MM2, RHO=0.0013, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					
2	444.	2 - 12MM	0.	1894.	YES	NO
	CRITICAL NEG MOMENT= 13.47 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 109.MM2, RHO=0.0010, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					
3	444.	2 - 12MM	2418.	4500.	NO	YES
	CRITICAL NEG MOMENT= 12.84 KN-MET AT 4500.MM, LOAD 108					
	REQD STEEL= 104.MM2, RHO=0.0009, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					

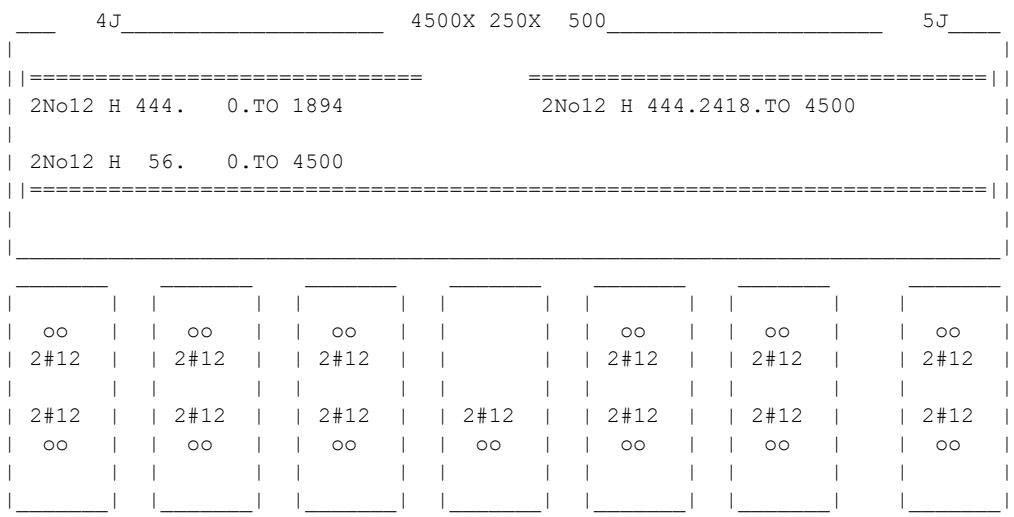
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	43./	83.	7./	13.	117/	109
375.	63./	60.	10./	10.	117/	109
750.	79./	39.	13./	6.	117/	109
1125.	92./	22.	15./	4.	117/	114
1500.	101./	9.	16./	1.	117/	114
1875.	107./	0.	17./	0.	117/	0
2250.	109./	0.	17./	0.	117/	0
2625.	107./	0.	17./	0.	117/	0

3000.	101./	5.	16./	1.	117/ 113
3375.	92./	18.	15./	3.	117/ 113
3750.	79./	35.	13./	6.	117/ 108
4125.	63./	56.	10./	9.	117/ 108
4500.	43./	80.	7./	13.	117/ 108

B E A M N O. 1 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 5.41 KNS Vc= 191.79 KNS Vs= 0.00 KNS
Tu= 1.71 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 7.48 KNS Vc= 106.31 KNS Vs= 0.00 KNS
Tu= 2.36 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 11 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	182.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     46.00 KN-MET  AT  2625.MM, LOAD  117|
|   REQD STEEL=   379.MM2, RHO=0.0034, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    41662.2 cm^4

  2        444.     3 - 12MM         0.          769.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     39.35 KN-MET  AT     0.MM, LOAD  117|
|   REQD STEEL=   321.MM2, RHO=0.0029, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   69. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    36333.1 cm^4

  3        444.     3 - 12MM      3168.         4500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     35.54 KN-MET  AT  4500.MM, LOAD  111|
|   REQD STEEL=   290.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   69. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    36333.1 cm^4

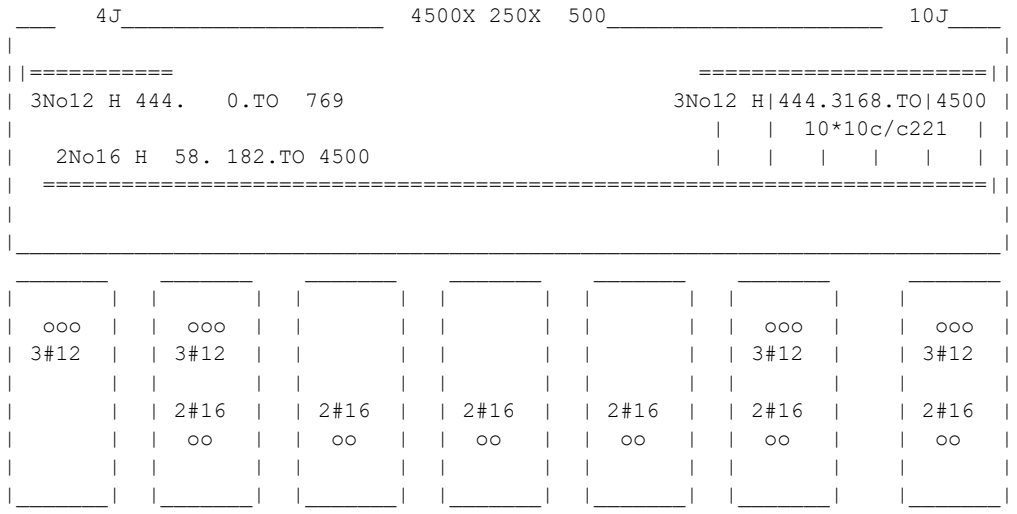

REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )         (SQ.  MM )                (KNS-MET )

    0.           0./     246.           0./       39.        0/  117
  375.           0./     103.           0./       17.        0/  117
  750.          38./       0.           6./        0.      110/    0
 1125.         113./       0.          18./        0.      117/    0
 1500.         190./       0.          30./        0.      117/    0
 1875.         245./       0.          39./        0.      117/    0
 2250.         278./       0.          44./        0.      117/    0
 2625.         289./       0.          46./        0.      117/    0
 3000.         277./       0.          44./        0.      117/    0
 3375.         244./       0.          39./        0.      117/    0
 3750.         188./      23.          30./        4.      117/  116
 4125.         111./     111.          18./       18.      117/  111
 4500.          13./     222.           2./       36.      117/  111
```

B E A M N O. 11 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   31.82 KNS  Vc=  120.62 KNS  Vs=    0.00 KNS
 Tu=    1.71 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```

AT END SUPPORT - Vu= 44.99 KNS Vc= 118.25 KNS Vs= 0.00 KNS
Tu= 2.42 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1810. MM



ACI 318-11 BEAM NO. 25 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4313.	YES	NO

CRITICAL POS MOMENT= 28.14 KN-MET AT 2250.MM, LOAD 117
REQD STEEL= 187.MM2, RHO=0.0014, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 64443.6 cm^4

2	542.	2 - 16MM	0.	1276.	YES	NO

CRITICAL NEG MOMENT= 43.15 KN-MET AT 0.MM, LOAD 109
REQD STEEL= 288.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 64443.6 cm^4

3	542.	2 - 16MM	2662.	4500.	NO	YES

CRITICAL NEG MOMENT= 43.70 KN-MET AT 4500.MM, LOAD 108
REQD STEEL= 291.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

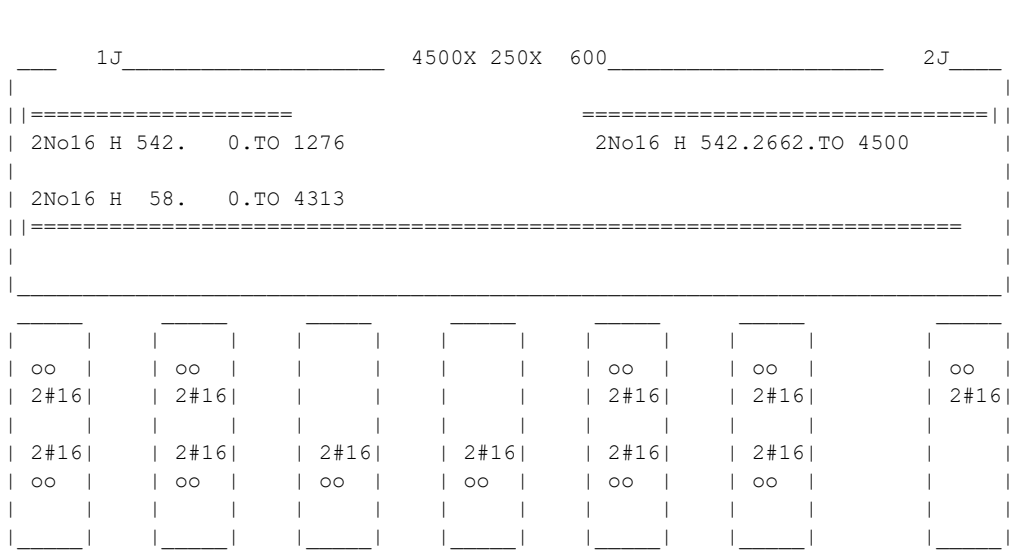
Cracked Moment of Inertia Iz at above location = 64443.6 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 219.	0./ 43.	0/ 109
375.	18./ 126.	4./ 25.	113/ 109
750.	53./ 49.	11./ 10.	108/ 114
1125.	90./ 0.	18./ 0.	108/ 0
1500.	113./ 0.	22./ 0.	108/ 0
1875.	134./ 0.	26./ 0.	117/ 0
2250.	142./ 0.	28./ 0.	117/ 0
2625.	134./ 0.	26./ 0.	117/ 0
3000.	116./ 0.	23./ 0.	109/ 0
3375.	93./ 1.	18./ 0.	109/ 113
3750.	56./ 52.	11./ 10.	109/ 113
4125.	21./ 129.	4./ 26.	114/ 108
4500.	0./ 222.	0./ 44.	0/ 108

B E A M N O. 25 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 28.36 KNS Vc= 133.72 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 40.08 KNS Vc= 139.38 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 5.0 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



ACI 318-11 BEAM NO. 30 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	759.	4303.	NO	NO
	CRITICAL POS MOMENT= 26.74 KN-MET AT 2625.MM, LOAD 117					
	REQD STEEL= 177.MM2, RHO=0.0013, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
2	542.	2 - 16MM	0.	1651.	YES	NO
	CRITICAL NEG MOMENT= 49.58 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 331.MM2, RHO=0.0024, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
3	542.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     34.97 KN-MET  AT  4500.MM, LOAD  102|
|   REQD STEEL=   233.MM2, RHO=0.0017, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

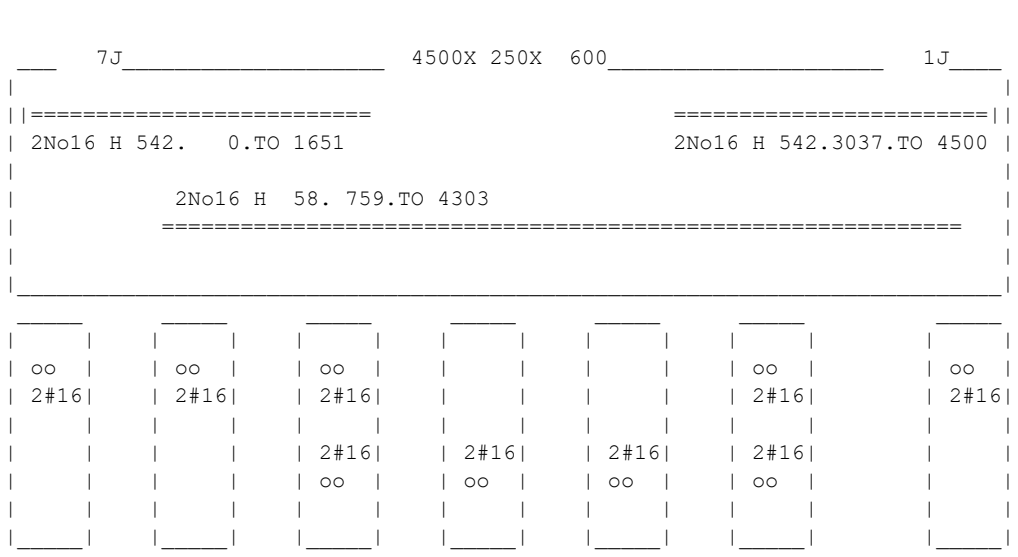
  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     252.           0./       50.        0/  117
  375.           0./     145.           0./       29.        0/  117
  750.           0./      62.           0./       12.        0/  110
 1125.          48./       9.          10./        2.      101/  115
 1500.          90./       0.          18./        0.      101/    0
 1875.         115./       0.          23./        0.      101/    0
 2250.         130./       0.          26./        0.      117/    0
 2625.         135./       0.          27./        0.      117/    0
 3000.         123./       0.          24./        0.      117/    0
 3375.          95./       0.          19./        0.      117/    0
 3750.          50./      14.          10./        3.      117/  108
 4125.          13./      85.           3./       17.      116/  102
 4500.           0./     177.           0./       35.        0/  102
```

B E A M N O. 30 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   28.36 KNS  Vc=  132.48 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   40.08 KNS  Vc=  132.48 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 584 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	759.	4303.	NO	NO
	CRITICAL POS MOMENT= 26.74 KN-MET AT 2625.MM, LOAD 117					
	REQD STEEL= 177.MM2, RHO=0.0013, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
2	542.	2 - 16MM	0.	1651.	YES	NO
	CRITICAL NEG MOMENT= 49.58 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 331.MM2, RHO=0.0024, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
3	542.	2 - 16MM	3037.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     34.97 KN-MET  AT  4500.MM, LOAD  102|
|   REQD STEEL=   233.MM2, RHO=0.0017, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

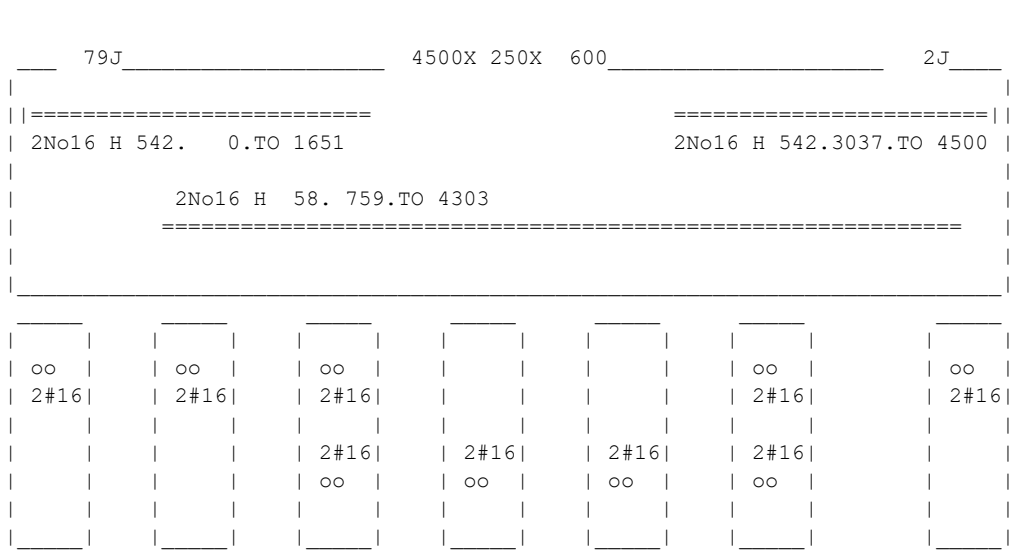
  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     252.           0./       50.        0/  117
  375.           0./     145.           0./       29.        0/  117
  750.           0./      62.           0./       12.        0/  110
 1125.          48./       9.          10./        2.      101/  115
 1500.          90./       0.          18./        0.      101/    0
 1875.         115./       0.          23./        0.      101/    0
 2250.         130./       0.          26./        0.      117/    0
 2625.         135./       0.          27./        0.      117/    0
 3000.         123./       0.          24./        0.      117/    0
 3375.          95./       0.          19./        0.      117/    0
 3750.          50./      14.          10./        3.      117/  108
 4125.          13./      85.           3./       17.      116/  102
 4500.           0./     177.           0./       35.        0/  102
```

B E A M N O. 584 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   28.36 KNS  Vc=  132.48 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   40.08 KNS  Vc=  132.48 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    4.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 587 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	182.	4500.	NO	YES
	CRITICAL POS MOMENT= 46.00 KN-MET AT 2625.MM, LOAD 117					
	REQD STEEL= 379.MM2, RHO=0.0034, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 41662.2 cm^4					
2	444.	3 - 12MM	0.	769.	YES	NO
	CRITICAL NEG MOMENT= 39.35 KN-MET AT 0.MM, LOAD 117					
	REQD STEEL= 321.MM2, RHO=0.0029, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 69. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 36333.1 cm^4					
3	444.	3 - 12MM	3168.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     35.54 KN-MET  AT  4500.MM, LOAD  111|
|   REQD STEEL=   290.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./   69. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                        |
|--------------------------------------------------------------|

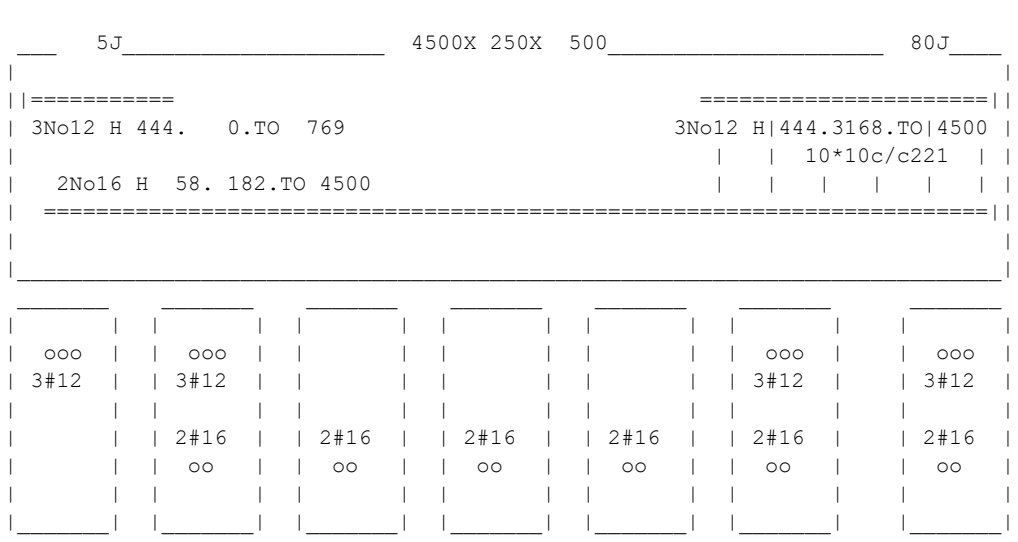
  Cracked Moment of Inertia Iz at above location =    36333.1 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     246.           0./       39.        0/  117
  375.           0./     103.           0./       17.        0/  117
  750.          38./       0.           6./        0.      110/    0
 1125.         113./       0.          18./        0.      117/    0
 1500.         190./       0.          30./        0.      117/    0
 1875.         245./       0.          39./        0.      117/    0
 2250.         278./       0.          44./        0.      117/    0
 2625.         289./       0.          46./        0.      117/    0
 3000.         277./       0.          44./        0.      117/    0
 3375.         244./       0.          39./        0.      117/    0
 3750.         188./      23.          30./        4.      117/  116
 4125.         111./     111.          18./       18.      117/  111
 4500.          13./     222.           2./       36.      117/  111
```

B E A M N O. 587 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   31.82 KNS  Vc=  120.62 KNS  Vs=    0.00 KNS
 Tu=    1.71 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   44.99 KNS  Vc=  118.25 KNS  Vs=    0.00 KNS
 Tu=    2.42 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   102
 NO STIRRUPS ARE REQUIRED FOR TORSION.
 REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
 PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM  C/C FOR 1810. MM
```



ACI 318-11 BEAM NO. 589 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	4313.	YES	NO
	CRITICAL POS MOMENT= 28.14 KN-MET AT 2250.MM, LOAD 117					
	REQD STEEL= 187.MM2, RHO=0.0014, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
2	542.	2 - 16MM	0.	1276.	YES	NO
	CRITICAL NEG MOMENT= 43.15 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 288.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS					
	REQD. DEVELOPMENT LENGTH = 404. MMS					
	Cracked Moment of Inertia Iz at above location = 64443.6 cm^4					
3	542.	2 - 16MM	2662.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     43.70 KN-MET  AT  4500.MM, LOAD  108|
|   REQD STEEL=   291.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS         |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                        |
|--------------------------------------------------------------|

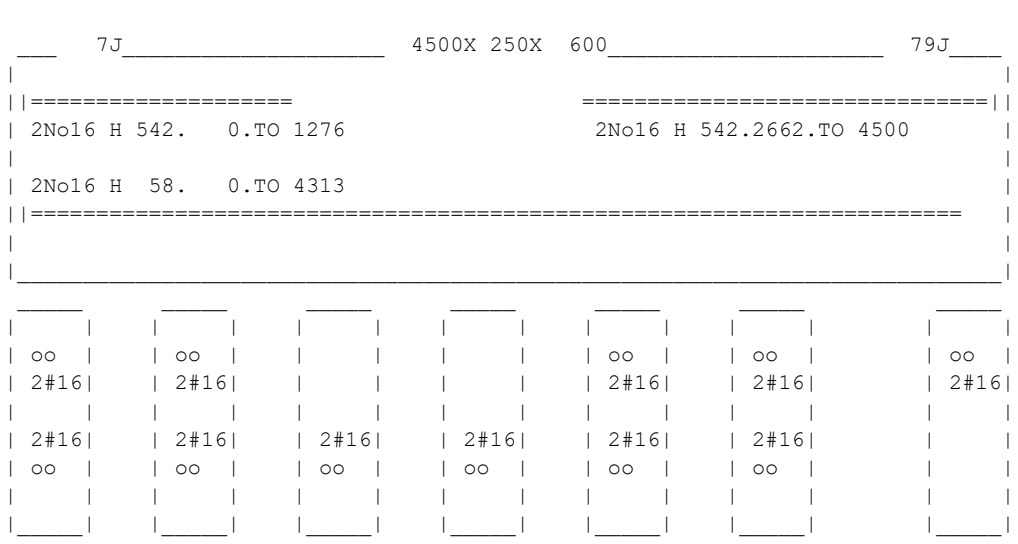
  Cracked Moment of Inertia Iz at above location =    64443.6 cm^4


REQUIRED REINF. STEEL SUMMARY :
-------------------------------
SECTION     REINF STEEL(+VE/-VE)     MOMENTS(+VE/-VE)    LOAD(+VE/-VE)
( MM )          (SQ.  MM )               (KNS-MET )

    0.           0./     219.           0./       43.        0/  109
  375.          18./     126.           4./       25.      113/  109
  750.          53./      49.          11./       10.      108/  114
 1125.          90./       0.          18./        0.      108/    0
 1500.         113./       0.          22./        0.      108/    0
 1875.         134./       0.          26./        0.      117/    0
 2250.         142./       0.          28./        0.      117/    0
 2625.         134./       0.          26./        0.      117/    0
 3000.         116./       0.          23./        0.      109/    0
 3375.          93./       1.          18./        0.      109/  113
 3750.          56./      52.          11./       10.      109/  113
 4125.          21./     129.           4./       26.      114/  108
 4500.           0./     222.           0./       44.        0/  108
```

B E A M N O. 589 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   28.36 KNS  Vc=  133.72 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   40.08 KNS  Vc=  139.38 KNS  Vs=    0.00 KNS
 Tu=    0.00 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```



ACI 318-11 BEAM NO. 590 DESIGN RESULTS

LEN - 4500. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	56.	2 - 12MM	0.	4500.	YES	YES
	CRITICAL POS MOMENT= 17.48 KN-MET AT 2250.MM, LOAD 117					
	REQD STEEL= 142.MM2, RHO=0.0013, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					
2	444.	2 - 12MM	0.	1894.	YES	NO
	CRITICAL NEG MOMENT= 13.47 KN-MET AT 0.MM, LOAD 109					
	REQD STEEL= 109.MM2, RHO=0.0010, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 37./ 138. MMS					
	REQD. DEVELOPMENT LENGTH = 305. MMS					
	Cracked Moment of Inertia Iz at above location = 25447.0 cm^4					
3	444.	2 - 12MM	2418.	4500.	NO	YES

```
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     12.84 KN-MET  AT  4500.MM, LOAD  108|
|   REQD STEEL=   104.MM2, RHO=0.0009, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   37./  138. MMS         |
|   REQD. DEVELOPMENT LENGTH =   305. MMS                       |
|--------------------------------------------------------------|

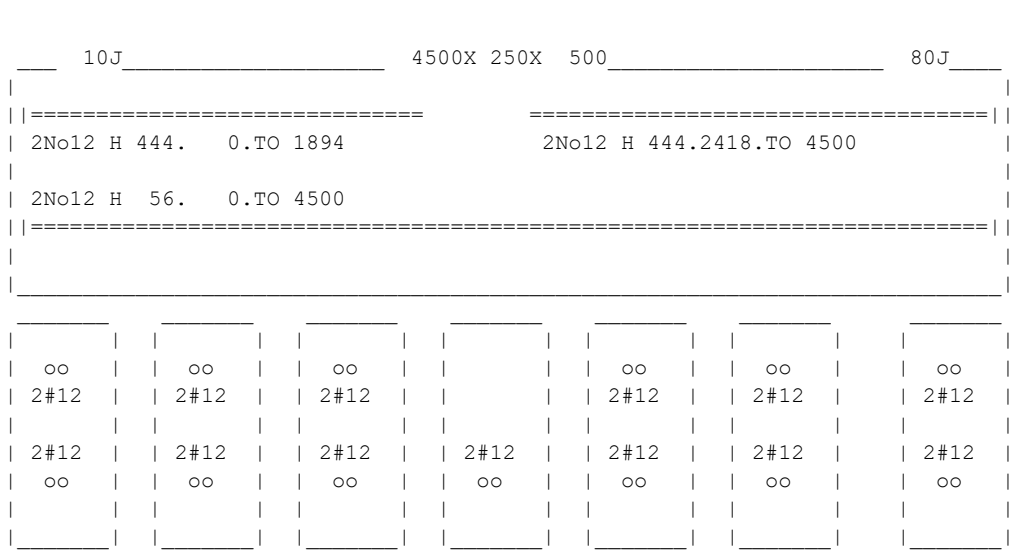
  Cracked Moment of Inertia Iz at above location =    25447.0 cm^4
```

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	43./ 83.	7./ 13.	117/ 109
375.	63./ 60.	10./ 10.	117/ 109
750.	79./ 39.	13./ 6.	117/ 109
1125.	92./ 22.	15./ 4.	117/ 114
1500.	101./ 9.	16./ 1.	117/ 114
1875.	107./ 0.	17./ 0.	117/ 0
2250.	109./ 0.	17./ 0.	117/ 0
2625.	107./ 0.	17./ 0.	117/ 0
3000.	101./ 5.	16./ 1.	117/ 113
3375.	92./ 18.	15./ 3.	117/ 113
3750.	79./ 35.	13./ 6.	117/ 108
4125.	63./ 56.	10./ 9.	117/ 108
4500.	43./ 80.	7./ 13.	117/ 108

B E A M N O. 590 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    5.41 KNS  Vc=  191.79 KNS  Vs=    0.00 KNS
Tu=    1.71 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    7.48 KNS  Vc=  106.31 KNS  Vs=    0.00 KNS
Tu=    2.36 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                   STIRRUPS ARE NOT REQUIRED.
```



**********END OF BEAM DESIGN**********

194. DESIGN COLUMN 3 4 8 31 32 34 583 585

ACI 318-11 COLUMN NO. 3 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

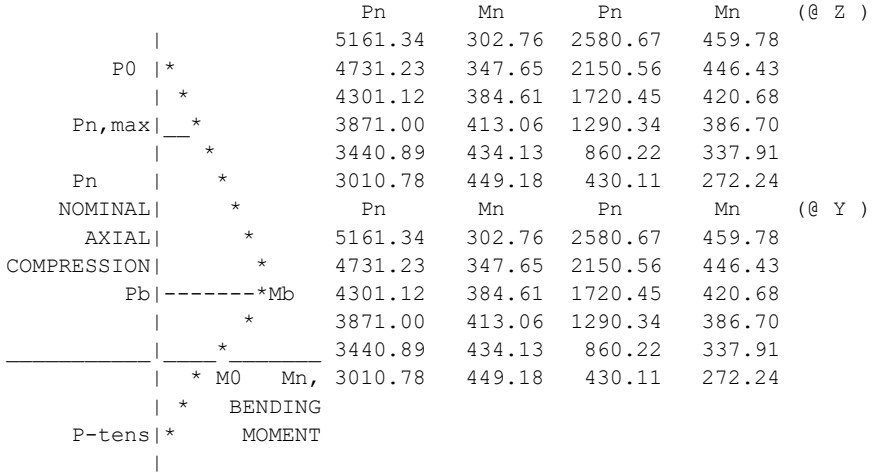
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	6.99	0.01870

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	45.36	0.12134



ACI 318-11 COLUMN NO. 4 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

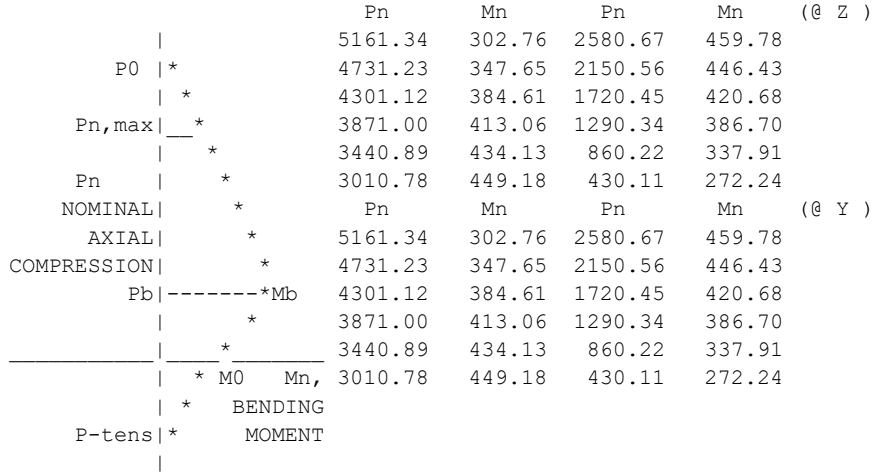
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	6.99	0.01870

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	45.36	0.12134



ACI 318-11 COLUMN NO. 8 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

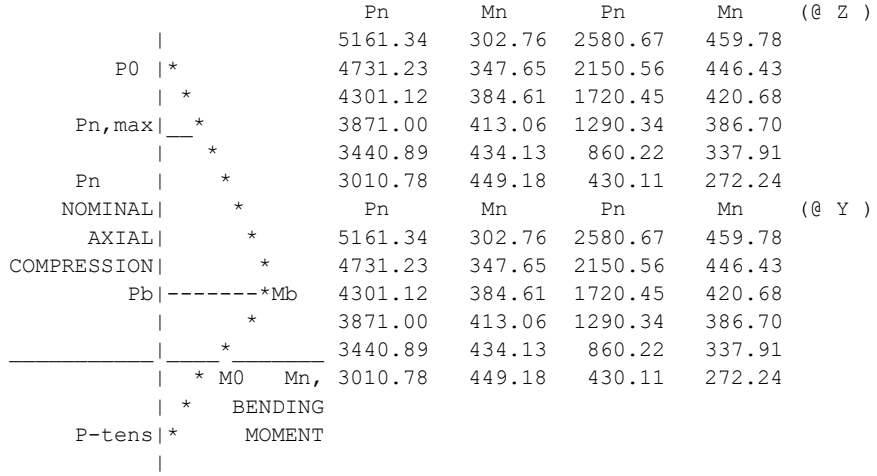
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	6.99	0.01870

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	45.36	0.12134



ACI 318-11 COLUMN NO. 31 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

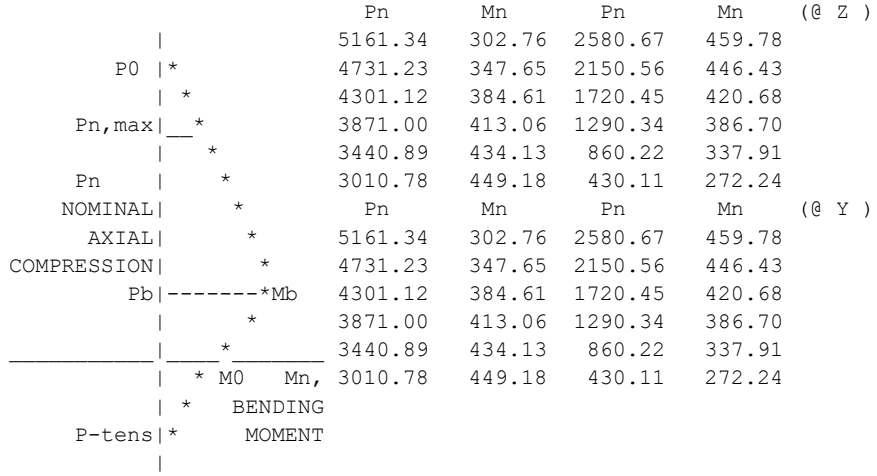
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	320.41	13.88	0.02166

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	320.41	10.22	0.01596



ACI 318-11 COLUMN NO. 32 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

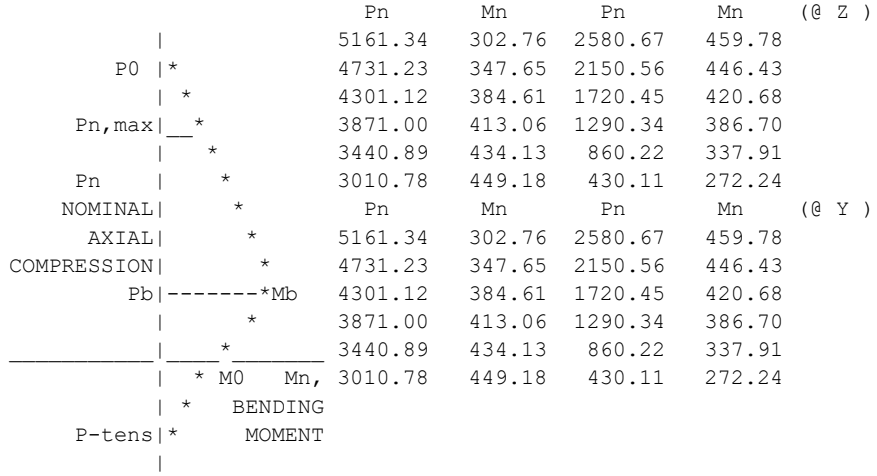
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	320.41	13.88	0.02166

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	320.41	10.22	0.01596



ACI 318-11 COLUMN NO. 34 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

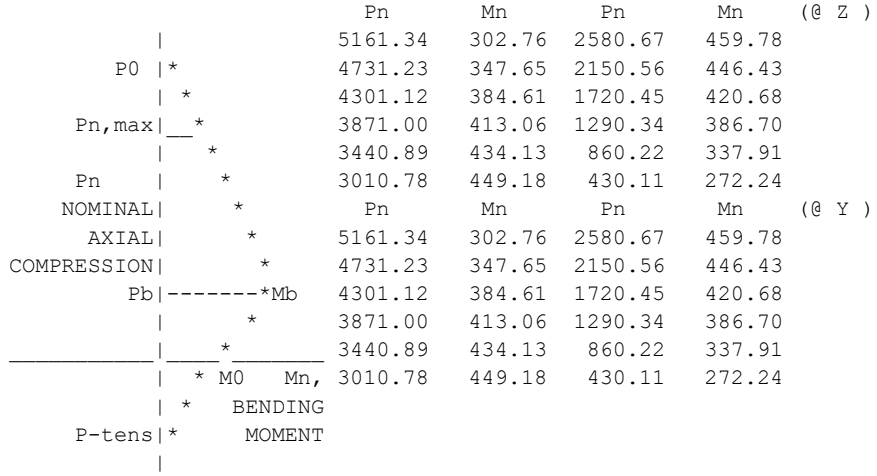
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
198.54	-1039.71	320.41	13.88	0.02166

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	**P-tens.**	**Des.Pn**	**Des.Mn**	**e/h**
198.54	-1039.71	320.41	10.22	0.01596



ACI 318-11 COLUMN NO. 583 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

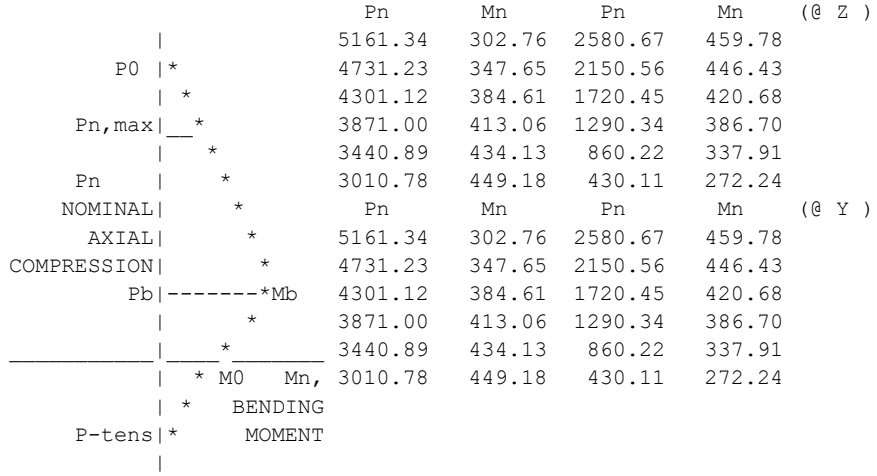
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	6.99	0.01869

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	98.40	45.36	0.12134



ACI 318-11 COLUMN NO. 585 DESIGN RESULTS

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.241	101	END	0.650

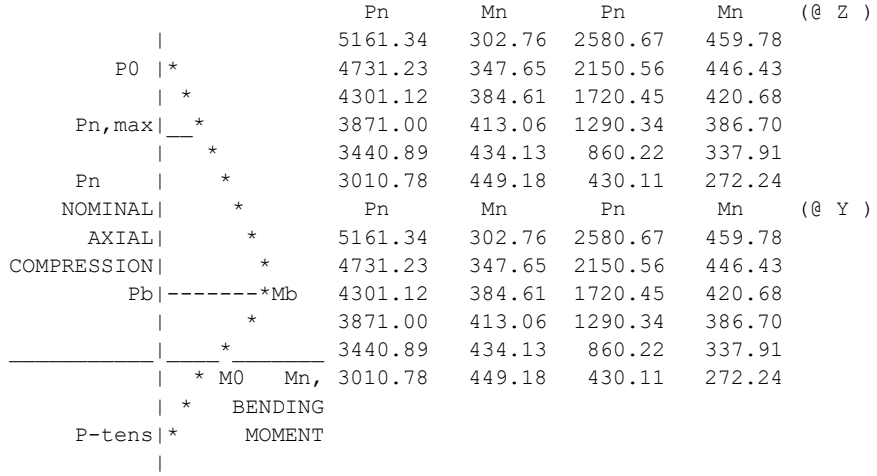
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	320.41	13.88	0.02166

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6989.31	5591.45	2495.29	461.50	184.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
198.54	-1039.71	320.41	10.22	0.01596



**********END OF COLUMN DESIGN RESULTS**********

195. CONCRETE TAKE
196. END CONCRETE DESIGN

********** CONCRETE TAKE OFF **********
(FOR BEAMS, COLUMNS AND PLATES DESIGNED ABOVE)

NOTE: CONCRETE QUANTITY REPRESENTS VOLUME OF CONCRETE IN BEAMS, COLUMNS, AND PLATES DESIGNED ABOVE.
REINFORCING STEEL QUANTITY REPRESENTS REINFORCING STEEL IN BEAMS AND COLUMNS DESIGNED ABOVE.
REINFORCING STEEL IN PLATES IS NOT INCLUDED IN THE REPORTED QUANTITY.

TOTAL VOLUME OF CONCRETE = 9.6 CU.METER

BAR SIZE NUMBER	WEIGHT (in lbs)
10	41
12	91
16	256
20	1007
*** TOTAL=	1394

197. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= OCT 16,2020 TIME= 21:56:28 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
* Copyright (c) 1997-2016 Bentley Systems, Inc. *
* http://www.bentley.com *
****